

For each of these circuits:

1. Calculate the total resistance in ohms,

a) Series connection

$$R_T = R_1 + R_2 + R_3$$

$$R_T = 20 + 30 + 50$$

$$R_T = 100\Omega$$

b) Parallel connection

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$\frac{1}{R_T} = \frac{8}{100}$$

$$R_T = 12.5 \Omega$$

2. Calculate the total current from the power supply.

a)

$$I_T = \frac{V}{R_T}$$

$$I_T = \frac{125}{100}$$

$$I_T = 1.25A$$

b)

$$I_T = \frac{V}{R_T}$$

$$I_T = \frac{125}{12.5}$$

$$I_T = 10A$$

3. Calculate the current through each resistor

a)

$$I_T = I_1 = I_2 = I_3$$

$$I_T = 1.25A$$

b)

$$20 \, \Omega = I = \frac{125}{20}$$
$$= 6.25A$$

$$100 \, \Omega = I = \frac{125}{100}$$
$$= 1.25A$$

$$50 \, \Omega = I = \frac{125}{50}$$
$$= 2.5A$$

4. Calculate the Voltage drop through each resistor

a) $V=IR$

$$\text{Through } 20\Omega: V = 1.25(20)$$
$$= 25V$$

$$\text{Through } 30\Omega: V = 1.25(30)$$
$$= 37.5V$$

$$\text{Through } 50\Omega: V = 1.25(50)$$
$$= 62.5V$$

b)

$$V_T = V_1 = V_2 = V_3$$
$$= 125V$$

5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)

a) $P= VI$

$$\text{Through } 20\Omega: P = 25(1.25)$$
$$= 31.25W$$

$$\begin{aligned}\text{Through } 30\Omega: P &= 37.5(1.25) \\ &= 46.875\text{W}\end{aligned}$$

$$\begin{aligned}\text{Through } 50\Omega: P &= 62.5(1.25) \\ &= 78.125\text{W}\end{aligned}$$

$$\begin{aligned}P_T &= (31.25 + 46.875 + 78.125)\text{W} \\ P_T &= 156.25 \text{ W}\end{aligned}$$

b)

$$\begin{aligned}\text{Through } 20\Omega: P &= 125(6.25) \\ &= 781.25\text{W}\end{aligned}$$

$$\begin{aligned}\text{Through } 100\Omega: P &= 125(1.25) \\ &= 156.25\text{W}\end{aligned}$$

$$\begin{aligned}\textit{Through } 50\Omega: P &= 125(2.5) \\ &= 312.5\text{W}\end{aligned}$$

$$\begin{aligned}P_T &= (781.25 + 156.25 + 312.5) \text{ W} \\ P_T &= 1250 \text{ W}\end{aligned}$$